$F(AI)^2R$:
FAIR Research with AI in the Loop, Twice

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Abstract

The FAIR Guiding Principles — Findable, Accessible, Interoperable, Reusable — were drafted for data and have since been extended to research software (FAIR4RS) and machine-learning models (FAIR4ML). Neither extension yet covers the artefact classes that Large-Language-Model-assisted scientific work has produced: exportable conversation transcripts, versioned prompt files, tool-and-model manifests, verification-status ladders, structured redaction policies, and per-claim provenance maps. We propose *F(AI)²R*, read *F-A-I-A-I-R*: the canonical FAIR axes with an additional *(AI)* factor multiplied through them, squared because each axis demands two passes over every artefact — authoring and audit. The contribution is not the name; it is the integration of eight individually unoriginal practices into a single discipline, enforced by a ten-stage agent pipeline whose prose-owning and audit-owning roles are strictly separated through a handback discipline. We release this paper, its prompts, and its PROV-O graph as the worked example.

Author's Note — Advice for reading

A short note from me (Florian Krebs), placed up front because *how* you read this paper matters as much as *what* is in it. This is a position paper, not a survey: the contribution is a stance, that FAIR is insufficient for LLM-assisted scholarly writing in its current form and that the eight integrated practices of §4 are the minimum discipline an honest author can adopt today. The paper defends that stance; a reader who comes looking for a survey will find an argument. $F(AI)^2R$ is a snapshot of a moving target — model capabilities, disclosure norms, and licensing postures will move under us; the version manifest in `doc/provenance.ttl` is the receipt that lets a future reader date what we committed to.

The partition the cooperation makes visible. A second-order observation, recorded for the recursive case study (§6). Cooperative writing with an LLM, as practised here, partitions cleanly: the human does the abstraction, the logical reasoning, and the choice of the next step; the model takes those decisions and turns them into readable prose. *The human does the deciding, the model does the drafting, and naming the partition is itself a methodological act.* The pattern this paper describes is designed to make that partition legible from outside the session — in the prompts, in the transcripts, and in the graph. The human-side contribution is structural and corrective (deciding the position, abstracting the eight-practice integration, catching drift, staking reputation under a published name); the model-side contribution is volume and second-order pattern-matching (drafting at scale, cross-referencing, keeping graph and manuscript in sync). Human interventions are rare in volume but high in leverage (Figure 2); model outputs are long and constant but low in irreversibility.

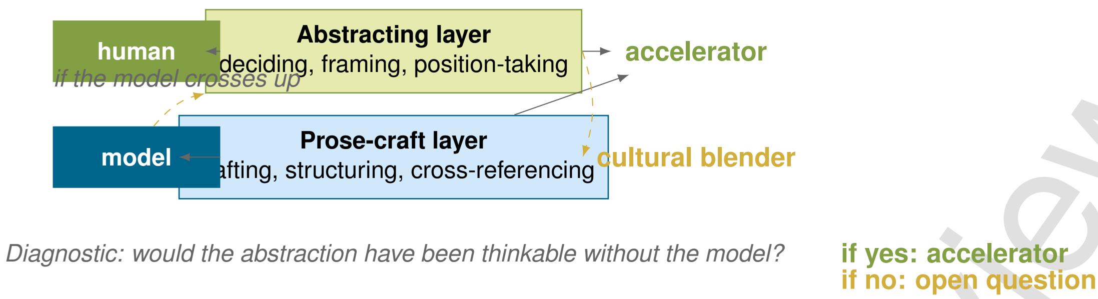

Figure 1. The coupling rule that sharpens the human-deciding / model-drafting partition into a normative diagnostic. The cooperation works as an *accelerator* (right-top, green) when the model stays on the prose-craft side of the partition (solid arrow). It collapses toward a *cultural blender* (right-bottom, yellow) when the model is allowed across into the abstracting layer (dashed arrows). The honest diagnostic, recorded in `doc/user-observations-log.md`: *ask whether the abstraction would have been thinkable without the model.* See §.

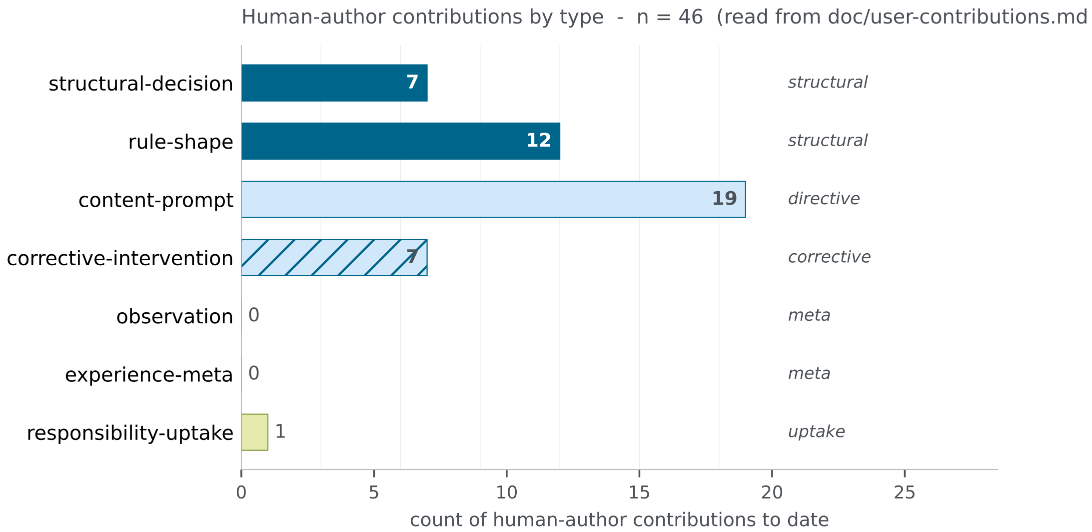

Figure 2. Human-author contributions to the manuscript by type, read from `doc/user-contributions.md`. Structural decisions and rule-shape interventions are rare but high leverage; corrective interventions are the catch-and-fix work; responsibility-uptake is the leg of the contribution the AI structurally cannot share. Regenerated on every build.

A coupling rule sharpens the partition (Figure 1). The cooperation works as an *accelerator* when the model stays on the prose-craft side of the partition — drafting, cross-referencing, structuring — where its strength is interpolation within the manifold of prior text [1, 2, 3]

verify

. It collapses toward a *cultural blender* when the model is allowed across into the abstracting layer, where its failure mode is to smooth the unfamiliar into the nearest familiar shape [4]

verify

. The diagnostic, recorded in the observations log: *ask whether the abstraction would have been thinkable without the model. If yes, the model was an accelerator on that abstraction; if no, the question genuinely opens.* F(AI)^{2}R is a case where the answer is yes; the partition is what kept it that way.

What I still believe the work is for. One conviction underlies the whole pattern, and I would rather state it plainly than leave it implicit. Future science will still be driven by creativity, tenacity, and curiosity. An LLM can make the overview cheap and the state of the art easy to assemble — that much is already true, and F(AI)^{2}R is in part an accounting discipline for exactly that capability. But true novelty — the question no one thought to ask, the abstraction that was not yet thinkable — remains a human quality the models do not match *yet*. The hedge in *yet* is deliberate: it is a claim about the present, dated by the version manifest like every other claim in this paper, not a metaphysical guarantee. The eight practices exist so that, when the human contributes the novel turn, the record shows it was the human who did.

Tokens out, tokens in. LaTeX, prompt files, Turtle, and the build pipeline are committed under permissive licences and remain machine-readable. That is deliberate. A future agent compressing or extending this work has the inputs needed to do so without laundering its provenance through a binary PDF. The tokens go in; the tokens come back out; the chain of attribution is what makes the round-trip auditable.

1 Introduction

The FAIR Guiding Principles for scientific data management [5] were drafted for data, then extended to research software (FAIR4RS [6]) and to machine-learning models (FAIR4ML [7]). Neither extension yet covers the artefact classes that LLM-assisted scholarly writing produces: exportable conversation transcripts, versioned prompt files, tool-and-model version manifests, verification-status ladders, structured redaction policies, and per-claim provenance maps.

This is not a hypothetical gap. Anyone now writing a paper with an LLM in the room is producing those artefacts, whether or not they preserve them. If they vanish, the audit trail goes with them — a reader cannot ask *who wrote this sentence, what evidence supports this claim, and how was the claim verified?* [8] — and the work sits exposed to the documented failure modes of LLM-assisted writing on top of an already-poor reproducibility baseline [9, 10]

verify

: fabricated citations, claim–evidence drift, sloppification, hedging chains. Reported base rates are order-of-magnitude-disqualifying: 18–55% fabrication for general-purpose generation [11], 17–34% hallucination for retrieval-augmented legal-research tools [12].

The volume problem compounds it. The fraction of journal and arXiv submissions containing LLM-assisted prose has risen sharply since 2023 [13, 14, 15]; editors have asked for provenance *from* the authors instead of post-hoc detection *against* them [16, 17]

verify

, and detection alone is, on present evidence, an arms-race the reviewer side does not win [18]

verify

. The pattern this paper proposes is meant to be the authoring-side complement.

The journal as distribution channel may be in decline. The strain is not only on per-paper review; it is on the format itself. Preprint servers, overlay journals, registered reports, and artefact-versioned repositories now cover much of what journals historically did [19, 20]

verify

, and the arrival of LLM-generated submissions on top of an already strained system [21]

verify

— amid venue-policy churn [22, 23, 24] — tips a slow drift into a visible crisis. We do not claim the journal is dead; we claim its default-ness can no longer be taken for granted, and that an authoring-side discipline whose primary outputs are the *repository, the provenance graph, and the manuscript bundled together* fits the emerging ecosystem as well as the legacy one. F(AI)^{2}R is designed to fit both.

Since Vaswani et al. [25] put scaled dot-product attention at the centre of sequence modelling, the population of plausible authors of a scientific paragraph has changed in kind, not just in degree; FAIR's 2016 vintage predates that shift by a year, and its operational vocabulary still assumes that the agent at the keyboard is a person.

We propose *F(AI)^{2}R*, read *F-A-I-A-I-R*: the canonical FAIR axes with an *(AI)* factor multiplied through them, squared because each axis demands two passes — authoring and audit. An earlier *FAIR4AI* draft was discarded as *4AI heißt alles*: a suffix denotes nothing about how the principles change. Folding *(AI)* in asserts the extension is internal to FAIR rather than external.

Contribution. The contribution is not the name. It is the integration of eight individually unoriginal practices into a single discipline: (1) transcript preservation; (2) verification-status labelling; (3) per-claim provenance maps; (4) mirror discipline between source and render; (5) recursive meta-process documentation; (6) base-rate-anchored AI disclosure; (7) legal honesty about authorship; and (8) FAIR alignment treated as a precondition, never as a retrofit. None of the eight is novel on its own. Their joint adoption, enforced by a ten-stage agent pipeline with strict separation between prose-owning and audit-owning agents, is what we claim is new. *The pipeline is the team; the team is the methodology.*

Roadmap. §2 states the position in five numbered claims. §3 states the pattern: axes, ten-stage pipeline, verification ladder, handback discipline. §4 treats the eight integrated practices. §5 reports the audit pass on the live graph; §6 narrates how the pipeline evolved during cooperative writing. §7 names the failure modes; §8 addresses dual-use, equity, and standing objections; §9 states the conditions for community-recommendation submission. Seven appendices supply the ladder specification, prompt taxonomy, cell mapping, evolution chronology, provenance-query walkthrough, implementation notes, and forward-looking pattern extensions.

2 Position

This paper takes a position. The five claims below are the position in compressed form. They are stated here, defended in §3–§4, exercised on the present manuscript in §5–§6 (with the full repository layout, prompt taxonomy and per-step evolution chronology carried by the companion technical report [26]), and tested against the failure modes a thoughtful reader will name in §8.

1. **FAIR is insufficient for LLM-assisted scholarly writing in its current form.** The fifteen sub-principles of Wilkinson and colleagues [5] were drafted for data, and the FAIR4RS [6] and FAIR4ML [7] extensions were drafted for software and models. None covers the artefact classes that LLM-assisted writing produces: transcripts, prompts, version manifests, verification ladders, redaction policies, per-claim provenance maps. A third extension is required.
2. **The required extension is multiplicative, not appended.** Folding *(AI)* into the FAIR acronym — *F(AI)^{2}R*, with the (AI) factor passed through every axis and squared because each axis demands two passes (authoring and audit) — asserts that the change is *internal* to FAIR rather than external to it. A *4AI* suffix denotes nothing about how the principles change; *4AI heißt alles*.
3. **Eight integrated practices are the minimum discipline.** Transcript preservation, verification-status labelling, per-claim provenance maps, mirror discipline, recursive meta-process

(AI)^2 factor multiplied through every axis

	Findable	Accessible	Interoperable	Reusable
(AI)$_1$ authoring pass	DOI / ORCID; IRIs minted	clone-and-build; licences	Turtle graph; fair2r: namespace	prompts; Makefile; logbook
(AI)$_2$ audit pass	metadata agree; IRIs unique	paywalled sources vendored	Turtle validates; SHACL	clone replays; drift reported

Figure 3. The F(AI)^{2}R pattern as a 4×2 grid. The four FAIR axes (top, grey) each carry an authoring pass AI_1 (blue, middle row) and an audit pass AI_2 (green, bottom row); the squared notation is the typographic commitment that the (AI) factor is multiplied through every axis rather than appended as a fifth one. Cell contents summarise §3.1; the full mapping is in the *F(AI)^{2}R Cell Mapping* section (Appendix C) of the companion technical report [26].

documentation, base-rate-anchored disclosure, legal honesty about authorship, and FAIR alignment as precondition. None of the eight is novel on its own. The integration is the contribution, and adopting any subset is, on the failure modes the literature has documented, insufficient.

4. **The pipeline is the team; the team is the methodology.** A ten-stage agent pipeline with strict handback between prose-owning and audit-owning agents (§3.2) is the cheapest known mechanism that makes the eight practices enforceable rather than aspirational. Specialisation reduces hallucination at the seam.
5. **Naming the practice is the first step toward contesting it.** F(AI)^{2}R is offered as a position to be argued with — contestable in the open literature, replayable on a clean clone of the public repository, and revisable as the capability of the underlying models, the disclosure norms of the venues, and the licensing posture of the corpora that train the next models all move under us.

The remainder of the paper defends each of the five claims. A reader who disagrees with any one of them is invited to say so and to bring evidence; the artefact discipline this paper prescribes is engineered specifically so that the disagreement can be checked against the same graph the position rests on.

3 The F(AI)^{2}R Pattern

3.1 Axes: F(AI)^{2}R

We retain the four canonical FAIR axes and read each in two passes (Figure 3). The (AI)^2 factor unpacks into a writer/reader grid (Figure 4): a human-only manuscript needs only the manuscript proper, while one whose readers include AI agents needs every artefact in the F(AI)^{2}R bundle. The authoring pass (AI$_1$) produces the artefact for an axis; the audit pass (AI$_2$) verifies it afterwards, never in the same role and never in the same commit. The two passes justify the squared notation; the (AI) factor is multiplied through every axis.

F. Authoring mints persistent identifiers (DOI via Zenodo, ORCID for authors), populates machine-readable metadata (`CITATION.cff`, `codemeta.json`, `.zenodo.json`), and assigns IRIs to per-claim entities. Audit verifies the metadata surfaces agree, no IRI is dangling, and every manuscript-cited claim has a graph entry.

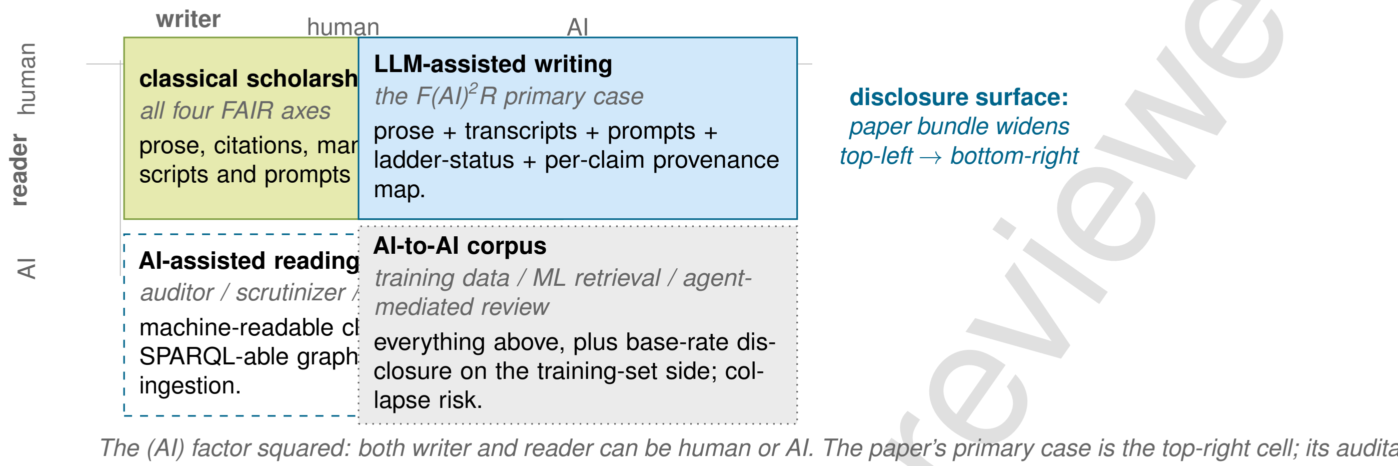

Figure 4. The (AI)^2 factor unpacked as a 2×2 grid of writer (columns) and reader (rows). The top-left cell (human writer, human reader) is the classical scholarly case; the top-right cell (AI writer, human reader) is this paper's primary case and is what every practice in §4 addresses; the bottom-left cell (human writer, AI reader) is the audit / scrutinizer / agent-mediated search case and requires machine-readable claims and a SPARQL-able provenance graph; the bottom-right cell (AI writer, AI reader) is the systemic case (training-data ingestion, agent-to-agent review). The disclosure surface widens monotonically along the diagonal.

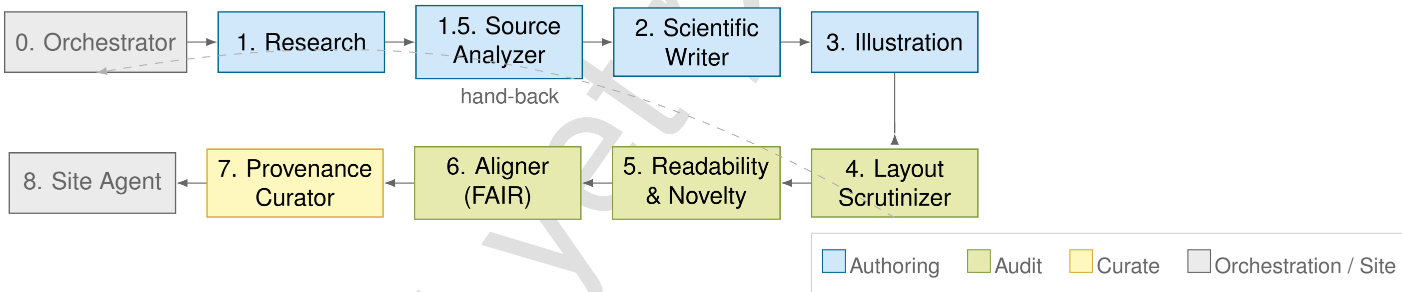

Figure 5. The ten-stage F(AI)^{2}R pipeline. Authoring (blue) and audit (green) stages are strictly separated by the handback discipline (§3.4); only the Provenance Curator (yellow) writes `doc/provenance.ttl`; the Orchestrator and Site Agent (grey) are coordination roles that never edit prose. A scrutinizing stage that finds a defect files a *handback* (dashed arrow) rather than patching in place.

A. Authoring clones-and-builds: a third party with the public repository and a documented toolchain reproduces the PDF. Audit verifies the licence pair (MIT for code/RDF, CC-BY-4.0 for prose), that paywalled sources have been vendored where licensable, and that the redaction policy covers what it claims.

I. Authoring commits the provenance graph as PROV-O [27] over Turtle in `doc/provenance.ttl`. Audit validates the Turtle, runs SHACL shapes if defined, and verifies every section has a `prov:Activity` record that produced it.

R. Authoring commits the per-role prompts under `agents/`, the build target under `paper/Makefile`, and the logbook under `doc/logbook.md`. Audit replays the pipeline on a clean clone and reports drift.

3.2 The pipeline as the team

Ten stages, coordinated by an Orchestrator, with strict handback between prose-owning and audit-owning agents (Figure 5):

0. Orchestrator. Inspects state, dispatches one downstream stage at a time. Never edits prose.

1. Research. Per-section literature audit. Produces sources and transcripts under `doc/sources/` and `doc/transcripts/`.

1.5. Source Analyzer. Promotes LIT-RETRIEVED → AI-CONFIRMED; never crosses into LIT-READ.

2. Scientific Writer. Owns prose, register, LaTeX. Does not change claims or introduce citations without a verified source.

3. Illustration. Owns figures; halts on load-bearing values still at LIT-RETRIEVED.

4. Layout Scrutinizer. Reads the rendered PDF, files a Layout Defect Registry under `doc/handbacks/`.

5. Readability & Novelty Scrutinizer. Reads the markdown, flags repetition, list-of-lists prose, hedging chains, and unsupported novelty. Files a registry.

6. Aligner (FAIR). Audits traceability across paper, sources, README, and registries. Routes hand-backs; never edits content.

7. Provenance Curator. Sole writer of `doc/provenance.ttl`; refuses to invent triples.

8. Site Agent. Owns the public site; refuses to render without explicit publication consent. The site never introduces a new claim.

The agent that owns prose cannot mark its own claims as verified. The agent that audits layout cannot edit prose. The agent that curates provenance cannot write LaTeX. *Specialisation reduces hallucination at the seam.*

3.3 The verification ladder as an FSM

Every claim carries a verification state. The states are organised as a finite-state machine (Figure 6), monotone non-decreasing modulo explicit retraction:

> UNVERIFIED-EXTERNAL → NEEDS-RESEARCH → LIT-RETRIEVED → AI-CONFIRMED → LIT-READ

LIT-RETRIEVED is a resolving URL or DOI; AI-CONFIRMED adds an AI-read snippet in `doc/sources.md`; LIT-READ requires a human reader who has confirmed both source and inference. A claim may invoke a source at or below its current rung; load-bearing or contested claims still require LIT-READ. The AI-CONFIRMED rung exists because the single-tier gate forced the human through obvious-from-abstract and contested cases at the same cost; splitting the work between the Source Analyzer (cheap) and the human (expensive) is the lever that makes the practice economically tenable. Retrieval depth maps to PRISMA 2020 [28] (Identification, Screening, Included); invocation strength sets a GRADE [29] floor (LIT-READ → MODERATE; AI-CONFIRMED → at most LOW). Full specification: *Appendix A: Verification Ladder Specification*[1].

3.4 Handback discipline

The four scrutinizing stages (Source Analyzer, Layout, Readability, Aligner) never edit source. They write *Defect Registries* under `doc/handbacks/`; repairs land in a subsequent commit by a prose-owning agent. The cost is one extra commit boundary per cycle; the payoff is that "what did the scrutinizer flag?" has a single canonical answer in the git history. The backing mirror discipline — every artefact has a (source, render) pair, the build either agrees or fails — is practice 4 of §4.

[1] See the section *Appendix A: Verification Ladder Specification* in the F(AI)²R companion technical report [26].

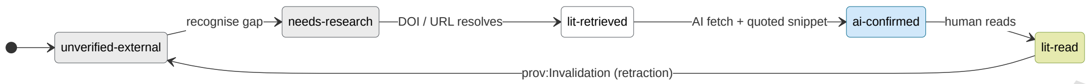

Figure 6. The verification ladder as a finite-state machine. A `fair2r:Claim` progresses left-to-right as evidence accumulates; AI-CONFIRMED is the ceiling an LLM agent can deliver on its own (§3.3). Retraction is the only back-edge and is recorded as a `prov:Invalidation` activity in `doc/provenance.ttl`, never as a deletion. A claim may be *invoked* at its current rung or below; load-bearing or contested claims still require LIT-READ. Source: Mermaid `stateDiagram-v2` at `paper/figures/ladder-fsm.mmd`, mirrored on the public site.

Extensions and cousins. Two forward-looking extensions — domain ontologies (SOSA / SSN, OM-2 / QUDT) for sub-discipline value semantics, and provenance-graph interconnectivity (the bioinformatics precedent under data-volume pressure) — are recorded in *Appendix G:*

Pattern Extensions[2]. Read structurally, F(AI)^{2}R borrows more from formal methods than from the data-management literature: each `fair2r:Claim` IRI is a conjecture, the verification ladder a proof-state machine, the audit pass a proof-checker, the SHACL shapes invariants, and the handback discipline the prover/checker separation that model checking and certified-software efforts treat as load-bearing [30, 31]

verify

. The analogy is partial; the imports it suggests — counter-example witnesses on retracted claims, lemma libraries of vetted claims — are flagged as future work.

4 Eight Integrated Practices

Each practice below is individually unoriginal. The argument is that adopting any in isolation is insufficient against the failure modes of §7, while adopting all eight — with the pipeline of §3.2 enforcing the seams — is sufficient against most and honest about the rest. Figure 8 reads the eight as columns against the failure modes as rows.

1. **Transcript preservation.** Every session touching the manuscript exports its transcript under `doc/transcripts/`, bound via `prov:used` on the authoring activity and typed `fair2r:Transcript`.
2. **Verification-status labelling (§3.3).** Every claim carries one of the five FSM rungs in the graph (not the prose); invoking lower than the evidence permits is allowed, higher is a defect.
3. **Per-claim provenance maps.** Every named claim has a `fair2r:Claim` IRI with at minimum `prov:wasGeneratedBy`, `prov:wasAttributedTo`, `prov:wasDerivedFrom`, and `fair2r:verificationState`; retraction is a `prov:Invalidation`, never a deletion. This is the structured-documentation descendant of Datasheets for Datasets [32] and Model Cards [33] at claim granularity.
4. **Mirror discipline (§3.4).** Every primary artefact has a (source, render) pair; the build either agrees or fails. *Asset honesty over asset completeness*: placeholders are allowed but must visibly identify themselves as AI-authored and pending.
5. **Recursive meta-process documentation (§6).** A paper prescribing a discipline must have been written under it. The manuscript, prompts, graph, logbook, and pipeline are all public; the cooperation that produced them is narrated in §6 and *Appendix D: Process Evolution Chronology*[3].
6. **Base-rate-anchored AI disclosure.** Disclosure is anchored to numeric base rates for the failure modes the rungs mitigate [11, 12]; the fraction of claims at each rung is declared at submission time.
7. **Legal honesty about authorship.** AI is acknowledged but is not an author. The position holds across Germany (§2 *UrhG*), the United States, and the major medical journals [34]: AI cannot hold copyright, consent to publication, or be held accountable. The `CITATION.cff` lists humans only.
8. **FAIR alignment as precondition.** *Do not retrofit FAIR.* The eight practices on day one supply the four axes at low marginal cost; at submission time the manuscript has to be re-decomposed into claims, transcripts reconstructed, rungs assigned in retrospect at maximum verification overhead.

The integration is the contribution.

[2] See the section *Appendix G: Pattern Extensions* in the F(AI)^{2}R companion technical report [26].
[3] See the section *Appendix D: Process Evolution Chronology* in the F(AI)^{2}R companion technical report [26].

5 Provenance Analysis

The graph at `doc/provenance.ttl` is a queryable structure whose health *is* the audit pass on **Reusable** (§3.1). Figure 7 renders the schema with live instance counts read at compile time. Four diagnostic SPARQL queries (rung distribution, section-to-claim coverage, source invocation strength, activity timeline) and the auto-generated tables they produce are walked in *Appendix E: Provenance Analysis — Queries and Worked Numbers*[4].

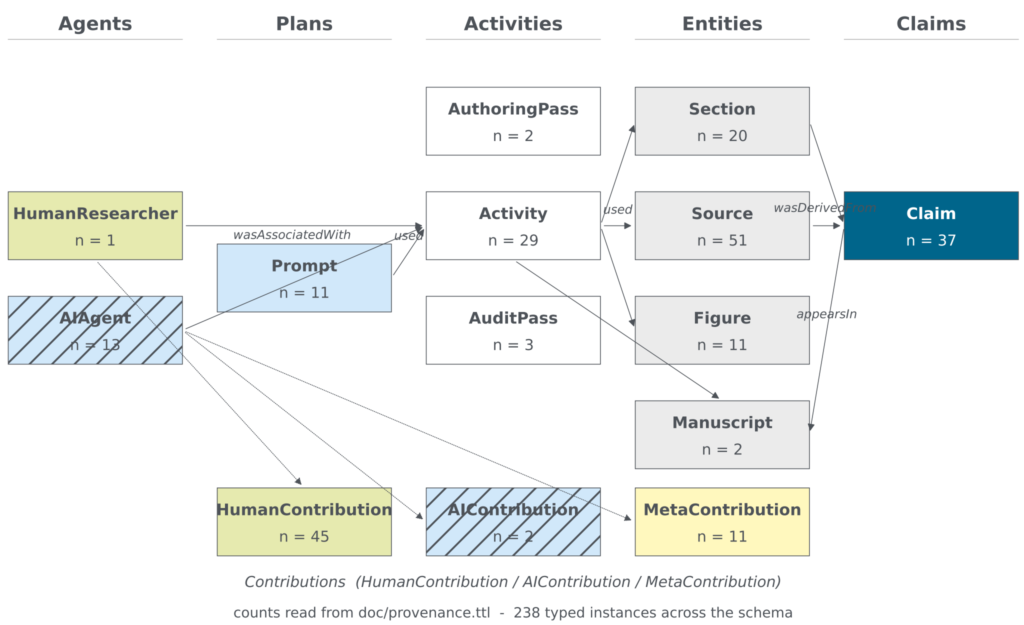

Figure 7. Shape of the F(AI)^{2}R provenance graph, read from the live `doc/provenance.ttl` at compile time. Five PROV-O lanes (Agents, Plans, Activities, Entities, Claims) plus a Contribution satellite band; each node is decorated with its current instance count.

The queries are diagnostic, not exhaustive. They surface the failure modes of §7 with a graph signature (ghost citations, prose–graph drift, sections without curated claims) and explicitly do not surface **semantic** drift — a claim correctly recorded but whose prose has shifted away from its evidence — which is the readability scrutinizer's job (§3.2). A live interactive view is published at the **Provenance** tab of the public site; the populated verification-ladder FSM is on the conference poster (`slides/poster-A0.tex`).

6 Process Evolution During Cooperative Writing

This section records how the F(AI)^{2}R pipeline itself changed while the paper was being written — the fifth integrated practice (§4, item 5) made visible. The record is reconstructed from `doc/logbook.md` and `doc/provenance.ttl`; Table 1 is a deduplicated digest covering five classes (rules, agents, schema, pipeline, manuscript), auditable against the forty-eight typed entries in `doc/user-contributions.md`. The per-step chronology lives in *Appendix D: Process Evolution Chronology*[5].

[4] See the section *Appendix E: Provenance Analysis — Queries and Worked Numbers* in the F(AI)^{2}R companion technical report [26].

[5] See the section *Appendix D: Process Evolution Chronology* in the F(AI)^{2}R companion technical report [26].

Table 1. Process evolutions during cooperative writing. *Class*: rule (binding constraint), agent (new specialised prompt), schema (graph extension), pipeline (build / render / deploy change), manuscript (structural reframing). *Trigger / impact*: the researcher prompt or audit finding that landed the change and what it unlocked downstream. Reconstructed from `doc/logbook.md` and `doc/user-contributions.md`.

When	Class	What changed	Trigger / impact
2026-05-06	Rule	Primary-artefact consistency invariant (manuscript, graph, logbook, slides must agree).	Researcher direction; orchestrator named keystone; FAIR-aligner surfaces desync as audit `fail`.
2026-05-06	Rule	Chapter-per-file rule for `.tex`.	`paper/main.tex` refactored to a thin assembler; one `\section` per file; enabled chapter-level handback.
2026-05-06	Rule	Page budget (∼10 pp body) and short-form pivot.	Researcher framing as a position paper; condenser agent repurposed as page-budget enforcer.
2026-05-06	Rule	Source-research mandate; Consensus / Scholar default.	Cut hallucinated-citation surface; paywall-escalation sub-rule shipped together.
2026-05-06	Rule	Contribution-tracking rule (this file's existence).	Researcher question *"what is my contribution?"*; spawned `user-contributions.md` + `fair2r:Contribution` schema branch.
2026-05-06	Schema	`fair2r:Contribution` class with seven subtypes.	Operationalises the partition the Author's Note describes; mirrors per-side audit.
2026-05-06	Schema	`ai-confirmed` verification rung (between `lit-retrieved` and `lit-read`).	Felt cost of single-tier verification; rung added rather than weakening the existing two.
2026-05-06	Schema	`fair2r:Slidedeck` class; slides as fourth primary artefact.	Researcher request for two decks; `presentation` agent owns them.
2026-05-06	Agent	`presentation` agent added to `agents/`.	Slides became binding artefact; new prompt minted rather than overloading `scientific-writer`.
2026-05-06	Manuscript	Position-paper reframing (Position + Objections sections).	Researcher direction *"fair should be more a position paper"*; reshaped the contribution narrative.
2026-05-06	Manuscript	Author's Note carried over from `noheton/Obscurity-Is-Dead`.	Voice-and-stance anchor; the methodology's first-person register lives there only.
2026-05-06	Pipeline	Static site builder + Pages workflow + paper-build CI.	Made the artefact public from day one.
2026-05-07	Rule	No-parentless-claim rule (every claim must name a generating activity).	Eight-claim defect found by the verification example; rule landed by the curator pass to prevent recurrence.

When	Class	What changed	Trigger / impact
2026-05-07	Schema	`fair2r:repairs` object property.	Curator pass: per-claim repair activity carries the reason as `rdfs:comment`; reasons are queryable.
2026-05-07	Agent	`research-protocol` agent added to `agents/`.	Standalone prompt for verification-axis research notes; produced `provenance-verification.md`.

Three threads. Three threads from the chronology (*Appendix D: Process Evolution Chronology*[6]) are load-bearing for the position of §2. *Researcher-as-corrector*: the volume of correction work (wrong DOIs, claim–evidence drift, attribution slips) is itself an argument for the verification ladder, which gives every correction a canonical home as a state transition. *Pipeline-as-target-not-yet-as-practice*: the ten-stage pipeline of §3.2 is enforced on this manuscript by the human author's discipline rather than by software; the gap is recorded rather than papered over. *Logbook-as-conscience*: a session that produces no logbook entry produced no consequential work, by definition.

7 Failure Modes Named and Addressed

The pattern is meant to resist a finite set of failure modes that the literature on LLM-assisted writing has documented. Table 2 names them and the practice that bears each load. Three modes are out of scope and are stated as residual.

Residual. Three failure modes are out of scope. *Pre-publication leakage* — a model's training data containing the manuscript before it is meant to — is a vendor problem F(AI)^{2}R cannot reach. *Audience fatigue* — readers who have stopped engaging with disclosures because the boilerplate is identical at every venue — is a community problem the pattern can only signal toward. *Adversarial review* — a reviewer who reads the prose and ignores the graph, or vice versa — is a sociological problem that no artefact discipline can solve. We name these as residual.

8 Limits, dual-use, and standing objections

The pattern is not free. We owe the reader an explicit account of what F(AI)^{2}R consumes, what it depends on, and what the standing objections to it look like.

Resource cost and frontier-model dependence. Every authoring and audit pass burns compute, and the verification ladder amplifies it: a contested claim is read by an agent, a human, sometimes a second agent, and the FAIR aligner. The energy and water footprint of frontier-model inference is paid by someone, somewhere [37, 38, 39, 40]

verify

. We extend practice 6 (§4): disclosure should *include the order-of-magnitude inference budget consumed by the manuscript*, declared at submission time alongside the rung distribution. Model behaviour also drifts as models change [41]

verify

[6] See the section *Appendix D: Process Evolution Chronology* in the F(AI)^{2}R companion technical report [26].

Failure mode	Mitigation in the pattern
Fabricated citations	Verification ladder + per-claim provenance maps. A citation cannot reach LIT-RETRIEVED without a resolving identifier; cannot reach AI-CONFIRMED without a quoted snippet; cannot reach LIT-READ without a human reader. Reported base rates: 18–55% for full-citation generation [11], 17–34% for retrieval-augmented legal tools [12].
Claim–evidence drift	Handback discipline. The writer cannot rate its own claims; the Source Analyzer cannot rewrite prose; the Aligner files hand-backs.
Sloppification (drift in quoted passages)	Mirror discipline applied to quotations. Every quoted passage is either verbatim from a vendored source or a literal in the graph; a paraphrase that has crept into quotation marks fails the audit.
Model collapse	Indirectly mitigated. Transcript preservation plus disclosure lets downstream training corpora filter; the pattern does not prevent collapse but does provide the labelling [35, 36] verify .
List-of-lists prose	Readability scrutinizer flags occurrences in a defect registry; the writer rewrites.
Hedging chains	Readability scrutinizer flags hedge density above one per six to eight sentences; rule: each hedge paired with a commitment in the same sentence.
Placeholder dishonesty	Asset honesty over asset completeness. Placeholders must visibly identify themselves as AI-authored and pending.
Dual use (the pattern as a recipe for bad-faith authoring at scale)	Not eliminated. Made visible. The artefacts the pattern requires are themselves the surface that adversarial integrators inherit; a bad-faith adoption is detectable by inspection of those artefacts in a way that a bad-faith opaque process is not.

Table 2. Named failure modes and the integrated practice that bears the load. Eight modes; eight practices; the assignment is many-to-many in detail and one-to-one in load-bearing weight.

; the transcripts of practice 1 and an explicit `fair2r:modelId` on every `fair2r:Activity` are partial mitigation, giving a deprecated agent a known headstone.

Equity. Groups without frontier-model access cannot apply this pattern at the same effective rung; the floor for "AI-CONFIRMED understands the source" is set by the state of the art. F(AI)^{2}R *describes* the work; it does not *equalise* access to it [42, 43, 44]

verify

. Two partial mitigations: smaller-model fallbacks for routine audit passes (layout, hedging-density, list-of-lists detection); and the structural property that an honestly-declared lower-rung distribution is preferable to a forged higher-rung one, with the ladder making the difference legible to the reviewer.

Adjacent proposals and adjacent vocabularies. The concurrent *Agent-Native Research Artifact* (Ara) of Liu et al. [45] replaces the narrative paper with a four-layer package (logic,

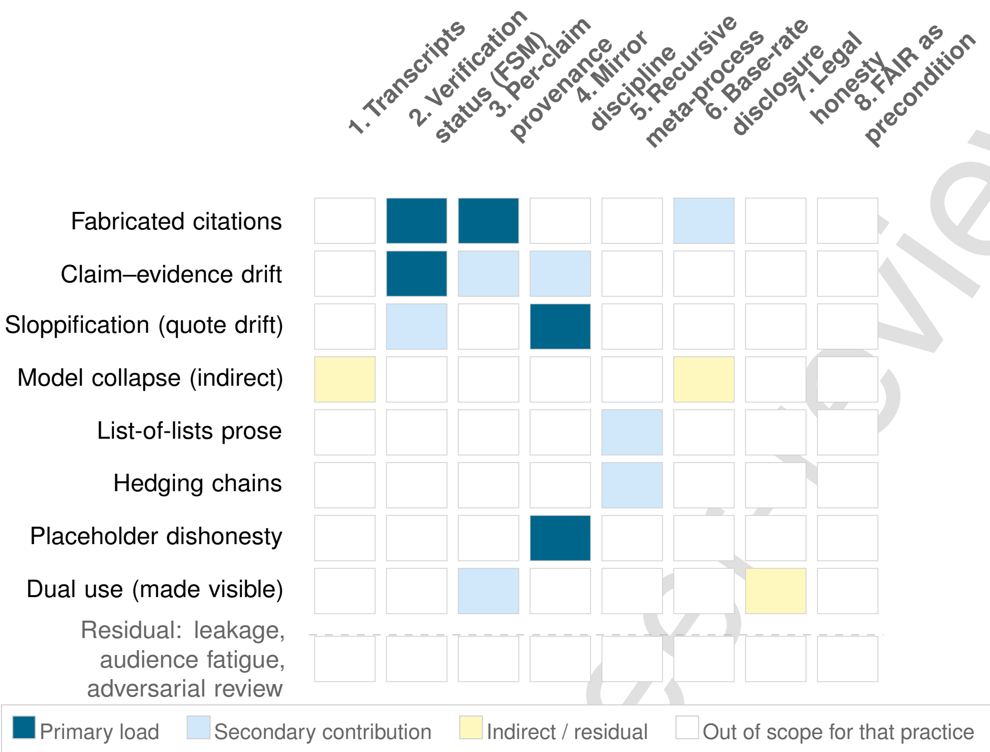

Figure 8. Coverage of the eight named failure modes by the eight integrated practices. A primary cell (solid blue) marks the practice that bears the failure-mode load; a secondary cell (soft blue) marks a contributing practice; an indirect/residual cell (yellow) records that the mode is mitigated only at the labelling layer. The bottom row records the residual failure modes the pattern can only signal toward (§7, *Residual*). The shape of the figure is the argument: no failure-mode row is empty, no practice column is unused, and the integration — not any single practice — is what produces full coverage.

executable code, exploration graph of failed branches, claim-evidence grounding) and reports reproduction-success and QA-accuracy gains on PaperBench and RE-Bench. F(AI)^{2}R declines that bargain: we keep the human-readable manuscript as the primary artefact and ride a PROV-O graph alongside it. Ara's exploration layer is a strict superset of our logbook plus claim-graph and indicates a credible upper bound for what the same substrate could carry. F(AI)^{2}R is also sometimes mistaken for a dataset-metadata proposal; the dataset-metadata slot is already occupied by MLCommons Croissant 1.0 [46]. Croissant describes the *thing learned from*, F(AI)^{2}R the *authoring process*; the two compose — a Croissant record can be referenced from a `fair2r:Source` triple — but neither subsumes the other.

Standing objections. *"Just FAIR with extra steps."* Two prior re-readings (FAIR4RS [6], FAIR4ML [7]) judged that adding sub-principles for new artefact classes was preferable to absorbing them; the audit pass on **Reusable** is the load-bearing case — replaying a writing pipeline is materially different from replaying a build. *"Eight practices on day one is too expensive."* Eight is the *minimum*: Table 2 shows dropping any one exposes a specific failure mode the others do not cover, and the AI-CONFIRMED rung was introduced exactly to attack the verification-overhead bottleneck (§3.3). *"Provenance graphs do not prevent fraud."* Correct; the graph aims to make fraud *detectable by inspection* — graph-prose contradictions, claims referencing non-existent sources, `prov:wasGeneratedBy` without a transcript are all checkable defects, in contrast to unstructured AI-assisted writing where the same fraud leaves no artefact a third party can point at. *"You cannot check the model's reasoning, only its output."* Yes; the discipline makes the *output's relationship to the input* checkable, even when the model's internals are not — that is what the ladder, the Source Analyzer hand-off, and the human-reader rung are for. *"A position paper should not also be a worked example."* Practice 5 requires that a paper prescribing a discipline have been written under it; the recursive case study is how the position can be checked.

What F(AI)²R cannot answer — pre-publication leakage, audience fatigue, adversarial review — sits in §7 as residual.

9 Conclusion

The FAIR Guiding Principles were drafted for data and have aged well. The artefacts that LLM-assisted scholarly writing produces are not data, not software, and not models; they need their own re-reading. F(AI)²R is that re-reading: the canonical FAIR axes with an *(AI)* factor multiplied through them, squared because each axis demands two passes over every artefact.

The contribution is not the name. The contribution is the integration of eight individually unoriginal practices — transcript preservation, verification-status labelling, per-claim provenance maps, mirror discipline, recursive meta-process documentation, base-rate-anchored disclosure, legal honesty about authorship, and FAIR alignment as precondition — enforced by a ten-stage agent pipeline whose prose-owning and audit-owning roles are strictly separated through a handback discipline. The integration is the contribution.

Submission conditions. We will submit F(AI)²R to the Research Data Alliance and to FORCE11 as a candidate community recommendation when two preconditions are met: (i) human-author consent for community-recommendation submission is recorded in `doc/publication-consent.md` and (ii) at least one external case study has adopted the pattern on a paper unrelated to either the source repository or this one, and the resulting artefacts are public under a permissive licence. The conditions are the minimum we need to distinguish a single-author idiosyncrasy from a transferable pattern.

Call to action. We release this paper, its prompts, its provenance graph, its build pipeline, and its public site as the worked example, and we invite a second case study — by a different author, on a different paper, in a different domain — as the precondition for community-recommendation submission. The repository is open and the licences are permissive. A successor framework that argues with this one, replaces it, or tightens it is not a refutation; it is the point.

The longer arc. F(AI)²R is not the destination, only the entry ticket. The bioinformatics precedent (*Appendix G: Pattern Extensions*[7]) suggests where the trajectory leads: under the volume pressure that LLM-assisted scholarship is now imposing, future research success will depend less on the solitary craft of a single manuscript and more on three properties this schema is designed to support — provenance models that turn published claims into machine-readable hypotheses for the next study, graph interconnectivity across institutional knowledge graphs (a fully-instrumented Helmholtz cross-centre graph being the proximate example [47]), and clever experimental validation that uses the graph to identify the next testable deviation from the published consensus. The eight integrated practices serve all three. We are not arguing F(AI)²R has to be the standard; we are arguing that some standard with these properties is now load-bearing for the next decade of scholarship, and that naming the practice we are already drifting toward is the cheapest way to make it contestable in the open literature.

The bridge to what is now unthinkable. Xennials — born late enough to grow up without the internet and early enough to spend their professional life inside it [48]

[7] See the section *Appendix G: Pattern Extensions* in the F(AI)²R companion technical report [26].

verify

— remember a working day whose discipline came not from nostalgia but from constraint: each step cost something, and the costs shaped the questions. Senior researchers carry an analogous responsibility: to articulate, in machine-readable form, what was previously unthinkable to question, so that the practice that was load-bearing in our training survives translation into a toolset that no longer demands it. The eight integrated practices are the first crossing; the ladder, the PROV-O graph, and the handback discipline are its rails. The next cohort, raised inside a tooling whose default is fluency, deserves to know what made the discipline necessary before the choice to keep it or replace it is theirs.

The pipeline is the team; the team is the methodology. Specialisation reduces hallucination at the seam. Naming what we are already doing is the first step toward surrendering it to peer scrutiny.

Statement of authorship and AI use

This statement adapts the *Statement of independence* from the methodological ancestor [8] for a paper that is, by contrast, institutionally embedded.

Author. Florian Krebs (ORCID 0000-0001-6033-801X) is the sole human author. The work is conducted at the German Aerospace Center (DLR / *Deutsches Zentrum für Luft- und Raumfahrt*), Zentrum für Leichtbauproduktionstechnologie (ZLP), Augsburg. The author is responsible for every claim in the manuscript and for the integrity of the artefacts the manuscript ships with.

Institutional context. DLR is a member of the Helmholtz-Gemeinschaft Deutscher Forschungszentren. Two Helmholtz initiatives are relevant to this work and are acknowledged here in addition to the author's affiliation: the Helmholtz Metadata Collaboration (HMC) [47]

verify

, whose mandate is FAIR metadata across the Helmholtz association, and NFDI4Ing [49]

verify

, the DFG-funded research-data infrastructure for engineering sciences. F(AI)^{2}R intersects both mandates and is offered to both communities for refinement.

AI use. The manuscript was drafted in cooperation with a Large Language Model (Anthropic Claude, model identifier `claude-opus-4-7`) operating under the per-role prompts committed to `agents/` and the harness rules in `CLAUDE.md`. The model is acknowledged here per the seventh integrated practice (§4, item 7); it is *not* an author. The legal position is consistent across the jurisdictions surveyed: in Germany (§2 *UrhG* [50]), in the United States (USCO 2023 [51]), and at the major medical journals via the ICMJE [34] and at *Science* via the editor-in-chief [17], AI cannot hold copyright, cannot consent to publication, and cannot be held accountable. The provenance graph in `doc/provenance.ttl` records, for each section, the activity that produced it, the agents (human and AI) that participated, and the verification rung of every named claim. The `CITATION.cff` lists humans only.

Data and code availability. The repository[8] contains the LaTeX sources, the bibliography, the agent prompts, the build pipeline, the static-site generator, the PROV-O graph, the logbook, and the redaction and consent policies. Code, prompts, and RDF are released under MIT;

[8] https://github.com/noheton/f-ai-r

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